

ThaiSLM v4 (final) — Architecture

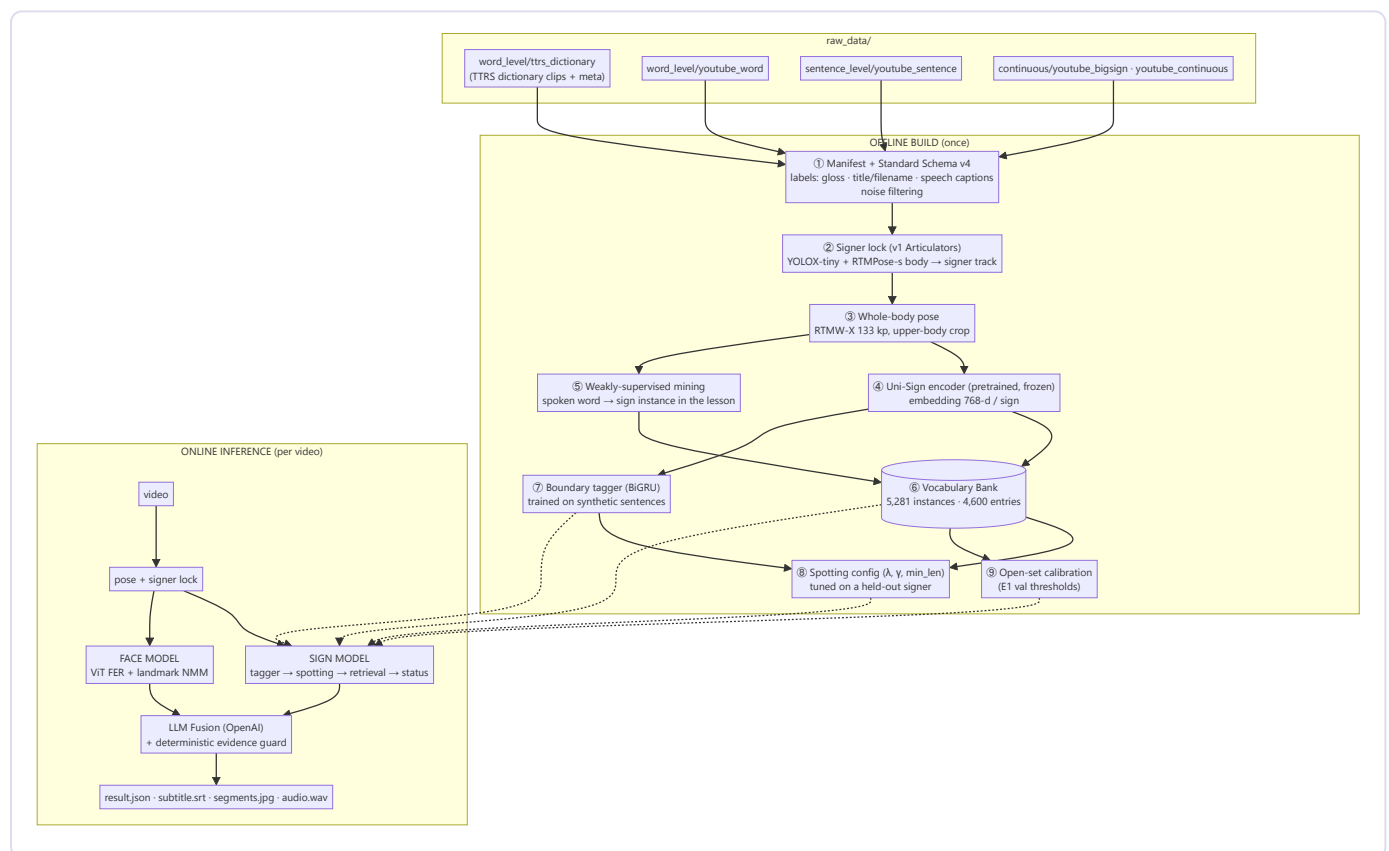
This document shows the path from `raw_data/` through the offline build to real-video inference. The configuration is the one used in the last run: `reports/v4_infer_final.json` and `notebooks/v4/07`.

Main idea: **the model is not a sentence classifier**. It turns video into pose, embeds each short sign segment with a pretrained encoder (Uni-Sign), and **retrieves** the nearest word from a vocabulary bank. When evidence is insufficient it outputs `[?] ≈ nearest word` instead of guessing.

Glossary of short labels used below:

- **E1**: new-signer vs full-dictionary retrieval evaluation.
- **OOV**: out of vocabulary.
- **NMM**: non-manual markers (brow, head, mouth).

1. Big picture



2. Offline build — step by step

① Raw data → manifest + Standard Schema v4

`modules/lexicon.py::build_schema_v4` · `scripts/v4.py schema`

```

raw_data/word_level/ttrs_dictionary/*.mp4 + meta → label = TTRS gloss (lemma, category, pos, synonyms, signer)
raw_data/word_level/youtube_word/*.mp4 + info.json → label = parsed from title (".../หมวดคำกริยาอาการ/ข้อบรรยาย" → ข้อบรรยาย)
raw_data/sentence_level/youtube_sentence/* → label = spoken words: th-orig VTT captions (word timestamps)
raw_data/continuous/* → excluded (broadcast/song: caption ≠ signing, lag, 2 people)

```

↓

```

artifacts/manifests_v4/clips_v4.parquet clip_id · source · subset · media_path · signer_id · title · label
label_source(gloss|title) · usable_for(vocab|weak_word|"" ) ·
exclude_reason
artifacts/manifests_v4/speech_words.parquet clip_id · word · t0 · t1 · source (7,468 hits, 819 dictionary words)
artifacts/manifests_v4/cleaning_report.json 171 clips dropped with a reason (no label 77, broadcast/song 47,
fingerspelling 40, TTRS >15 s 6, test entry 1)

```

② Signer lock (from v1) → artifacts/tracks/*.npz

modules/articulators.py

```

frames @12.5 fps → YOLOX-tiny person detection (every N frames)
→ RTMPose-s body 17 kp
→ signer lock (wrist confidence + size + layout zone, hysteresis) → body keypoints of the signer

```

③ Whole-body pose → artifacts/pose_v4/<clip>.npz (5,202 files)

modules/wholebody.py · scripts/v4.py pose --set ttrs|youtube_labelled · pose_test

```

signer body kp → upper_body_box (expanded so the hands get more pixels)
→ affine crop 256x192 → RTMW-X (rtmw-dw-x-1, ONNX Runtime CUDA, batch 48) → SimCC decode
npz: kp [T,133,2] (0..1) sc [T,133] fps hw t0
133 = body 17 · feet 6 · face 68 · left hand 21 · right hand 21

```

④ Encoder: Uni-Sign pose encoder (pretrained, frozen / zero-shot)

modules/unisign.py · checkpoint wlasl_pose_only_islr.pth (Uni-Sign, ICLR 2025)

```

kp [T,133,2], sc [T,133]
| upsample x2 (12.5 → 25 fps, the rate Uni-Sign was trained on)
| prepare_parts (official load_part_kp):
|   body [2T, 9,3] left [2T,21,3] (relative to wrist) right [2T,21,3] face [2T,18,3]
▼
per part: Linear(3→64) → spatial ST-GCN chain → temporal ST-GCN chain (left shares weights with right)
concat 4 parts (256×4) → Linear(1024→768) = "gcn" features [B,2T,768]
▼
[prefix embedding: "Translate sign language video to English: "] || gcn → mT5-base encoder (12 layers)
▼
"context" hidden states at frame positions [B,2T,768] → mean over valid frames → L2-norm → z ∈ R^768 (one sign)

```

Selected by a pilot on a 932-sign gallery: **WLASL ctx** R@1 0.23 / R@5 0.50. Ruled out:

- CSL checkpoint.
- ST-GCN-only features.
- The v1 DINOv2 RGB model (R@1 0.00).

⑤ Weakly-supervised mining (label = speech in the clip, no manual labels)

modules/mining.py · scripts/v4_build.py mine --rank_k 3

speech word w at time t (from `speech_words.parquet`)

- candidate windows in $[t-1.5\text{ s}, t+3.0\text{ s}]$ where the hands are raised
- embed each window (Ⓢ) and compare with the TTRS bank
- accept only if w is ranked top-3 among all ~4.6k words → 221 instances / 127 words / 56 videos

⑥ Vocabulary Bank → `artifacts/v4/bank.npz`

`scripts/v4_build.py embed_ttrs · embed_titles · bank`

```
instance-level bank
┌ Z [5281, 768] words source{ttrs 5036 · title 24 · mined 221} signer │
├ 4,600 entries: word 2,753 · compound 1,155 · phrase 655 · sentence 37 │
└ 412 entries have ≥ 2 signers │
└────────────────────────────────────────────────────────────────────────┘

score(query, entry) = mean of the top-2 cosine similarities among that entry's instances
```

The full list is in `vocab/corpus/vocab.json`; sentences and phrases are in `vocab/corpus/sentence.json`.

⑦ Boundary tagger (a segmentation helper) → `artifacts/v4/segmenter.pt`

`modules/segment.py · scripts/v4_build.py seg_train`

training data: synthetic sentences = 2-7 TTRS signs of the same signer
speed $\times 1-2.5$ · interpolated transitions · pauses · signing-space compression · one-hand dropout
input per frame: [ctx 768 || kinematics 18] (hand position/height/speed/angle/distance to face/confidence ...)
model: LayerNorm → Linear(786→192) → BiGRU $\times 2$ → Linear → {outside, begin, inside}
held-out signer: boundary F1 0.716

⑧ Spotting config → `artifacts/v4/spotting.json`

`scripts/v4_build.py spot_eval`

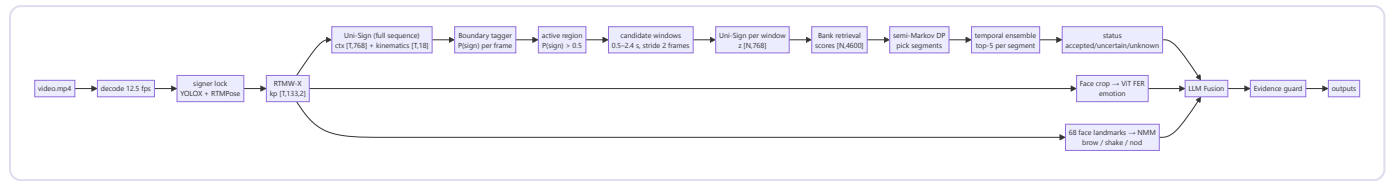
data: synthetic sentences of held-out signer `S_ttrs_00` · bank with ALL of that signer's clips removed (78% OOV)
grid search on 120 "tune" sentences → report on 120 "test" sentences
selected: $\lambda = 9$ · $\gamma = 1.0$ · `min_len` = 14 frames · windows 6-30 frames · stride 2 · candidate aggregation = sum
test: segment F1 0.93 · 5.1 predicted vs 4.8 true words per sentence

⑨ Open-set calibration → `artifacts/v4/calibration.json`

Quantiles of correct matches on E1 val (new signer): `accepted`: score ≥ 0.638 and margin ≥ 0.071 · `uncertain`: score ≥ 0.564 · otherwise `unknown`.

Tried and **not used** in final: metric adapter fine-tuning (worse on test; `config.json use_adapter=false`) · Whisper ASR labelling (not finished) · test-time augmentation consensus / DTW re-scoring (no gain).

3. Online inference — `modules/pipeline_v4.py::SignPipelineV4.run(video)`



3.1 Sign Model (detailed)

```
kp [T,133,2], sc [T,133]
```

- └(a) Tagger – sequence_features → Segmenter → prob [T,3] → act[t] = smooth(P_begin+P_inside) > 0.5
(on real video the tagger sees one long signing run with no internal boundaries → it only defines the active region)
- └(b) Candidate windows $W = \{[a,b] : a \text{ on a 2-frame grid, } 6 \leq b-a \leq 30, \geq 60\% \text{ of frames active}\}$ (~500-800 per 7 s)
- └(c) Retrieval per window: upsample → parts (body scale from the whole video) → Uni-Sign ctx → z
 $S[w, \text{entry}] = \text{mean top-2 cos}(z_w, \text{bank instances of entry})$
 $z\text{-score}(w) = (\text{top1} - \text{mean_vocab}) / \text{std_vocab}$ ← how much the best word stands out from the whole vocabulary
- └(d) Semi-Markov DP (spotting)
 maximise $\sum \text{picked} (z\text{-score}(w) - \lambda) - \gamma \cdot (\text{active frames not covered})$
 subject to non-overlapping windows, length $\geq \text{min_len}$ (14 frames ≈ 1.1 s)
 merge adjacent picks with the same best word
- └(e) Temporal ensemble per segment: candidates = $\sum$ top-5 scores of every window inside the segment (± 2 frames)
- └(f) Open-set status (per segment)
 accepted : score ≥ 0.638 and margin ≥ 0.071 and ensemble top-1 == best-window top-1 → "word"
 uncertain : score ≥ 0.564 → "word?"
 unknown : otherwise → "[?] $\approx$ nearest"

Per-segment output: {f0, f1, t0, t1, status, nearest, score, margin, z, candidates:[{word, score, support}x5], window_top1}

3.2 Face Model (separate from the Sign Model)

face box from face kp → ViT FER (trpakov/vit-face-expression), every 2nd frame, 224 px
 → emotion, probs (7 classes), strongest non-neutral, intensity

68 face landmarks + head kp (relative to the signer's own baseline)
 → brow_raise_tail (question?) · head_shake_cycles (negation) · head_nod_cycles · mouth_open · per-segment brow/mouth

3.3 Fusion + evidence guard (modules/translate.py)

```
payload = { segments: [t0,t1,status,nearest,top-5], face: {emotion, probs, nmm} }
|
▼ OpenAI (OPENAI_MODEL = gpt-4.1-mini, temperature 0, JSON mode), system rules:
|   use only each segment's candidates · unknown → [?] · uncertain only if the meaning is coherent
|   add ไหม/ไม่ from the face · no evidence → sentence_th = "" and confidence ≤ 0.3
▼
guard_violations : any chosen word not in that segment's candidates (0 so far)
evidence_gloss   : what the Sign model alone supports, e.g. "เพื่อน? | จมูกแบน? | [?]≈ไม่พอ | สะโพก"
EVIDENCE GUARD (deterministic code, does not trust the LLM):
  show sentence_th only if (≥ 1 accepted sign) AND (≥ half of the segments resolved)
  otherwise → sentence_th = "" · tentative_sentence_th = LLM text · meaning_th = "ตรวจพบท่ามือ N ช่วง ... ช่วง i ≈ word"
```

3.4 Outputs (result_reporting/inference_v4/<video>/)

file	content
result.json	segments + face + fusion + timings
subtitle.srt	one line per segment: word / word? / [?] ≈ word
segments.jpg	3 frames per segment + status + top-3
segment_prob.npy	tagger probabilities
audio.wav	MMS-TTS Thai (facebook/mms-tts-tha), only when a sentence passes the guard

4. Components: pretrained, trained, or rule-based?

component	type	source
YOLOX-tiny, RTMPose-s (signer lock)	pretrained, frozen	rtmlib
RTMW-X 133 kp	pretrained, frozen	rtmlib / MMPose
Uni-Sign ST-GCN + mT5-base	pretrained, frozen (zero-shot)	Uni-Sign WLASL ISLR checkpoint
Vocabulary bank	built from data (no gradient training)	TTRS + YouTube titles + speech mining
Boundary tagger (BiGRU)	trained (synthetic sentences)	this project
Spotting DP parameters, calibration thresholds	tuned on a held-out signer / E1 val	this project
ViT FER	pretrained, frozen	trpakov/vit-face-expression
NMM cues	rules on landmarks	this project
LLM fusion	API (gpt-4.1-mini) + prompt rules	OpenAI
Evidence guard	deterministic rules	this project
TTS	pretrained	facebook/mms-tts-tha

5. Where time goes (data_test, GPU)

stage	video 1 (9.6 s)	video 2 (6.6 s)
pose (signer lock + RTMW-X)	3.5 s	1.9 s
tagger	0.45 s	0.15 s
spotting + retrieval	4.2 s	3.1 s
face	0.5 s	0.2 s
LLM	3.8 s	1.8 s

6. Commands (build order)

```
conda activate hugging
python scripts/v4.py schema                # ① manifest + schema
python scripts/v4.py pose --set ttrs       # ②③ (uses v1 signer tracks)
python scripts/v4.py pose --set youtube_labelled
python scripts/v4_build.py embed_ttrs     # ④
python scripts/v4_build.py embed_titles
python scripts/v4_build.py mine --rank_k 3 # ⑤
python scripts/v4_build.py bank           # ⑥
python scripts/v4_build.py eval
python scripts/v4_build.py calib          # ⑨
python scripts/v4_build.py seg_train      # ⑦
python scripts/v4_build.py spot_eval      # ⑧
python scripts/v4_build.py infer --tag final --tts # inference on data_test/
python scripts/export_vocab.py             # vocab/corpus/vocab.json, sentence.json
```